LabGenius S4 Automated Material Handling Robot

User Manual



: Please read the entire user manual carefully before operating the equipment.

- Version：English Version V1.0
- Date：2024.11.05
- Variants：S4-1.5-0.4-Z0.4, S4-1.5-0.6-Z0.6, S4-1.5-0.6-Z0.8

Bioyond Robotics (Shanghai) Co., Ltd

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1. Preface

Welcome to the Bioyond Robotics LabGenius S4 Automated Material Handling Robot! Thank you for choosing Bioyond. We sincerely appreciate your trust in our products. This manual provides all the essential information needed to set up and operate the LabGenius S4 robot, with a focus on key instructions for its efficient use in a lab environment. Before you begin operating the system, please read through this manual thoroughly. Pay particular attention to the safety guidelines and precautionary measures to ensure safe and optimal use.

If you require assistance, our dedicated support team is here to help. Please don't hesitate to contact us with any questions or concerns.

1.1. Copyright Statement

Copyright Notice

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This document and its contents, including but not limited to text, images, and graphics, are protected by copyright and other intellectual property laws. No part of this document may be reproduced, distributed, or transmitted in any form or by any means, electronic or mechanical, without the prior written permission of Bioyond Robotics.

Disclaimer

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Product Information and Updates

Product specifications and software/hardware configurations may be subject to change without notice. For the latest information, please refer to the relevant product documentation or contact Bioyond Robotics.

Errors and Omissions

Bioyond Robotics strives to ensure the accuracy and completeness of this manual. However, errors or omissions may occur. If you notice any discrepancies or have questions, please reach out to our support team.

Modifications

Bioyond Robotics reserves the right to modify the product and this manual at its discretion without prior notice.

1.2. Version Information

Version	Issue data	History

V1.0	2024-12	First release

1.3. Scope of Application

The information provided in this manual is intended solely for the Bioyond Robotics automated material handling robot and its accompanying software and hardware components.

1.4. Warranty Information

Warranty

- Warranty Period: This equipment is warranted for ONE year from the date of purchase, as documented on the invoice.
- Warranty Coverage: The warranty covers defects in materials and workmanship under normal use.
- Warranty Exclusions:
  - Damages caused by improper use, abuse, accidents, or modifications.
  - Normal wear and tear, including cosmetic damage.
  - Consumables and accessories.
  - Damages caused by external factors such as natural disasters or power surges.
  - Damages caused by unauthorized repairs or modifications.
  - Delays in service requests.

Warranty Claims

- To make a warranty claim, please contact our customer support team.
- Provide the following information:
  - Equipment model number
  - Serial number
  - Purchase date
  - Detailed description of the problem

Maintenance

- Maintenance must be carried out by qualified technicians only.
- Failure to follow authorized maintenance procedures may void the warranty.

1.5. Contact Information

If you have any questions, please contact Customer Support Center of Bioyond Robotics.
All customers can contact us through the following websites:

https://www.bioyond.com/en

Please keep your serial number on hand for us to provide you the best service!

2. Safety

This chapter outlines the safety principles and guidelines for operating LabGenius S4 automated material handling robot. Integrators and users must thoroughly review this manual and strictly adhere to the content marked with warning signs.

2.1. Safety Information

2.1.1. Responsibility and Risk

This manual does not provide instructions on designing, installing or operating a robot system, nor does it address all peripheral equipment that may affect the safety of the robot system.
It is the responsibility of the user to ensure compliance with all relevant laws and regulations and to verify that no significant hazards are present in the application of the LabGenius S4.
Any safety information given in this manual must not be interpreted as a warranty by Bioyond Robotics that the equipment will not cause injury or damage even if all safety instructions are followed.
In particular, Bioyond Robotics does not accept liabilities for injury or damage caused for any of the following reasons:

- Use of the robot in ways other than its intended purpose.
- Incorrect operation or maintenance.
- Operation of the robot when safety devices are defective, improperly positioned or otherwise non-functional.
- Failure to follow the operation and maintenance instructions.
- Unauthorized design modifications
- Repairs on the robot and its spare parts carried out by inexperienced or unqualified personnel.
- Presence of foreign objects.
- Events categorized as force majeure.

2.1.2. Requirements on Personnel

Only personnel with appropriate training are authorized to install, maintain, service, repair, and use the LabGenius S4 automated materials handling robot. This includes but is not limited to electrical, mechanical, pneumatics, and other hazards identified in the risk assessment.
Individuals under the influence of alcohol, drugs or any other intoxicating substances are strictly prohibited from installing, maintaining, servicing, repairing, or operating the robot.
It is the responsibility of the plant operator to ensure that all personnel are adequately trained in the operation of the robot, as well as in responding to emergency or abnormal situations.

2.2. Safety Signals and Symbols

2.2.1. Signal

■ Safety signal：

Symbols	Designation	Significance
	WARNNING	Signal word used to indicate a potentially hazardous situation which, if not avoided, could result in serious injury.
	ELECTRICAL SHOCK	Signal word used to indicate a potentially hazardous situation related to electrical hazards which, if not avoided, could result in serious injury
	HOT SURFACE	Signal word used to indicate a potentially hazardous situation related to high heat/temperatures
	PINCH POINT	Signal word used to indicate a potentially hazardous situation related to motion
	ESD	Signal word used to indicate a potentially hazardous situation related to ESD(electrostatic discharge), which, if not avoided, could result in severe damage to the product
	CAUTION	Signal word used to indicate important facts and conditions

■ Signal for shipping：

Symbols	Designation	Significance
	UP	Signal word used to indicate package should be placed as the symbol
	STACK LAYER	Signal word used to indicate up to 4 layers of stacking
	KEEP DRY	Signal word used to indicate pay attention to moisture and water

2.3. Safety Precautions

When operating the robot, ensure the safety of the operator is the top priority. The following general precautions should be observed, and appropriate measures must be implemented to safeguard operators.



WARNING:

- Before powering up the robot for the first time, it is important to check whether the robot is complete, safe to operate, and free of damage. Ensure that all safety functions are functional, and safety guards are properly in place. Additionally, verify it complies with applicable national or regional safety production rules and regulations.
- The user must verify that all safety parameters and operation procedures are correct and that all safety functions are working properly. Each safety function must be inspected by qualified operators. The robot may only be powered on after it has undergone thorough and meticulous safety testing and achieved there quired safety level.
- Qualified professionals must install and calibrate the robot in strict accordance with the installation standards.
- After the robot has been installed, a comprehensive risk assessment must be conducted, and documentation must be retained.
- Safety parameters must only be set or modified by an authorized operator. Passwords or other protection measures must be implemented to prevent unauthorized individuals from accessing or altering safety parameters. After modifying safety parameters, the associated safety functions must be analyzed and verified.
- Press the emergency stop button to halt the robot in the event of an accident or abnormal operation
- A mechanical brake is installed in the robot joint in order to maintain the robot posture when the power is off. Frequent power cycling is not recommended.it is advised to maintain an interval of at least 10s between power off and on. Remain vigilant when the robot is powered on, even if it is stationary, it may be awaiting a signal to resume

	movement. ● The robot should not be placed near chemicals or equipment that generates strong magnetic field equipment. ● he LabGenius S4 robot integrates collision detection function. The robot automatically stops when the external force applied to it exceeds the user defined threshold. When using this function, the user must conduct additional risk assessments, especially if the robot installs end-effector. Collision detection should not serve as the sole safety protection mechanism.
	ELECTRICAL SHOCK ● Ensure the robot is installed in strict compliance with the requirements and specifications outlined in this manual ● Ensure the robot is properly connected to the protective grounding and residual current device (RCD) before powering on or starting any installation work ● Use the power cord provided with the appliance or one that complies with the standard. Avoid pinching or compressing power cord during use. Any power cord that is frayed, worn, or otherwise damaged must be replaced promptly. ● Do not plug in, assemble or dis-assemble during the system is powered on.
	HOT SURFACE ● Do not touch the robot while it is operating or immediately after it has stopped, as heat is generated during operation

2.4. Intended Use

The LabGenius S4 automated material handling robot is designed exclusively for use in laboratories or similar environments. It must not be used in applications outside its intended purpose, including but not limited to the following:

- Operation in flammable, explosive and other hazardous environments
- Incorporated into equipment for moving or carrying people or other animals
- Integration into medical equipment and other devices related to human life
- Use in devices with significant societal or public impact
- Deployment in vibration environments such as vehicles or ships
- Utilization as a climbing tool

2.5. Safety Assessment

The LabGenius S4 automated material handling robot can be combined with other equipment to form a complete system; however, the robot itself is not a standalone solution. As such, this manual does not cover all aspects of designing, installing, or operating a complete robot, nor does it address all safety concerns related to the peripherals of such a system. The safety of a complete robot installation depends on how the robot is integrated. It is the responsibility of the user or integrator to conduct a risk assessment of the design and installation of the complete system in compliance with the laws and regulations of the country in which it is located, as well as the safety codes and standards. Risk assessment is a critical tasks that integrators must complete., Integrators may refer to the following standards to implement the risk assessment process

- ISO 12100:2010 Safety of machinery - General principles for design - Risk assessment and risk reduction
- ISO 10218-2:2011 Robots for Industrial Environments, Safety Requirements for industrial robots - Part 2: Robot systems and integration
- RIA TR R15.306-2016 Technical Report on Industrial Robots and Robot Systems — Safety Requirements: Task-based Risk Assessment Methodology
- ANSI B11.0-2010 Safety of Machinery - General Requirements and Risk Assessment

Integrators of LabGenius S4 robot are required to fulfill, but are not limited to the following responsibilities:

- Conduct a comprehensive risk assessment of the complete robotic system
- Ensure that the entire system is accurately designed and properly installed
- Provide training to end users
- Develop a user manual for the complete system, including clear instructions for operation
- Establish appropriate safety measures
- Identify and eliminate hazards or reduce them to an acceptable level using appropriate methods during final installation

The risk assessment must consider all potential interactions between the operator and the robot during normal use and foreseeable misuse. The operator's neck, face and head must be protected from any contact. Operating a robot without peripheral safety guards requires a prior risk assessment to determine whether the associated hazards pose an unacceptable risk, for example

- Use of sharp end-effector or tools
- Handling of toxic or other hazardous substances
- Robot collisions with operators or objects
- Poor fixation of the robot or tools
- Impacts between the robot payload and solid surfaces.

The integrator must evaluate such hazards and their associated risk levels through comprehensive risk assessment. They must identify and implement appropriate measures to mitigate the risk to an acceptable level. It is important to note that other significant hazards may arise depending on specific robotic devices in use.

2.6. Safety Functions and Safety Data of LabGenius S4

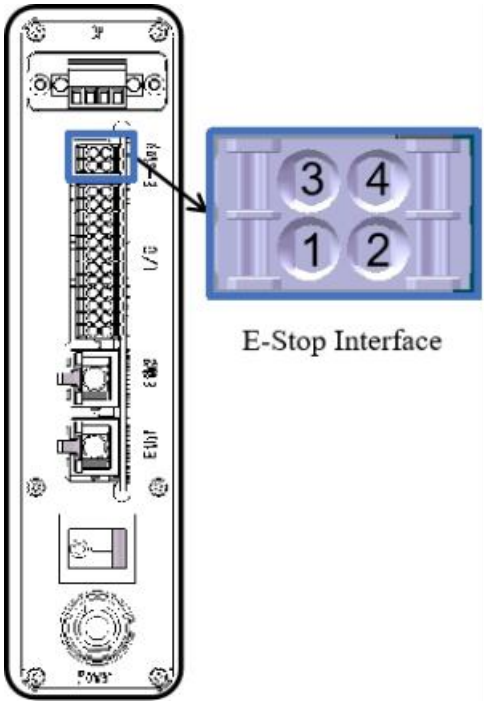
2.6.1. Emergency Stop

The LabGenius S4 automated material handling robot features an E-STOP interface designed for immediate stop of robot movement in the event of emergency. The E-STOP function complies with the category 1 stop as outlined by IEC60204-1.

The E-STOP interface meets the safety requirements of Category 3, PLd as specified in ISO13849-1. In addition, the emergency stops button, which is the user's responsibility, must conform to the requirements of IEC 60947-5-5. The safety data of the emergency stop safety interface is as follows:

Safety interface	MTTFd [Year]	DCavg [%]	PFHd [1/hour]
Estop channel 1	893	60	1.01*10^(-7)
Estop channel 2	893	60	1.01*10^(-7)

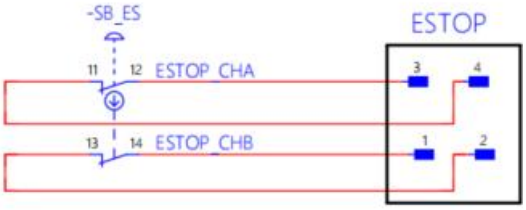
- Emergency stop interface description:



E-Stop Interface

Terminal	Signal	Description
1	ESTOP_CH1_P	E-STOP channel 1，positive
2	ESTOP_CH1_N	E-STOP channel 1，negative
3	ESTOP_CH2_P	E-STOP channel 2，positive
4	ESTOP_CH2_N	E-STOP channel 2，negative

- Connection explanation



2.6.2. Collision Detection

The LabGenius S4 automated material handling robot provides a collision protection function that complies with ISO/TS15066. It halts the robot's motion when a collision is detected, and the collision force exceeds the threshold. The collision detection stop function is classified as a category 2 stop, as specified in IEC60204-1.

2.6.3. Stop Category Description

The robot stop categories are defined in accordance with IEC 60204-1, as outlined in the table below

	Category 0	As defined in IEC 60204-1, stopping by immediate removal of power to the machine’s actuators. It is an uncontrolled stop
	Category 1	As defined in IEC 60204-1, a controlled stop with power available to the machine actuators to achieve the stop and then removal of power when the stop is achieved
	Category 2	As defined in IEC 60204-1, a controlled stop with power remaining available to the machine’s actuators.

2.6.4. Safe Stop Time and Safe Stop Distance

The safe stop time is defined as the duration between pressing the emergency stop button and the cessation of all robot motion. The safe stop distance refers to the distance the robot joint travels during the safe stop time.

Additionally, the emergency stop function of LabGenius S4 robot complies with a category 1 stop. The time from when the emergency stop is triggered to when the actuator power is cut off is 450ms.

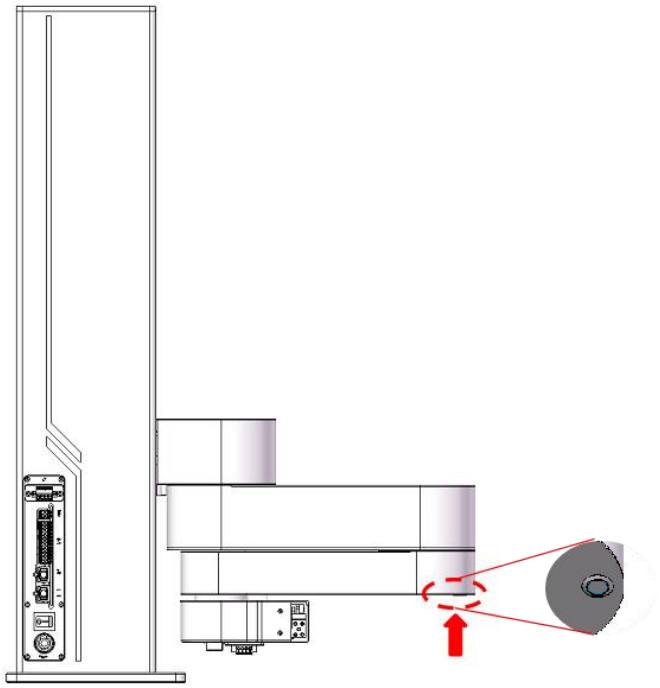
The following table shows the measured stop distance and stop time when emergency stop is triggered. These values correspond to the following configuration:

- Extension:100% (600mm, robot fully unfolds horizontally)
- Speed: 100%
- Payload: 1.5kg

Axis	Maximum speed	Maximum stop time	Maximum stop distance
Axis 1	500mm/s	200ms	50mm
Axis 2	1046mm/s	200ms	105mm
Axis 3	942mm/s	200ms	95mm

2.7. Movement Without Drive Power

All horizontal joints of LabGenius S4 are not equipped with brakes. They may be freely moved by the operator. The motor for vertical joints, however, features a mechanical brake. The operator can press the brake release switch, located beneath the inner link, to disengage the Z axis brake without drive power during an E-STOP or in abnormal situations.



: Ensure that you securely hold the cantilever while pressing the brake release button.

: The brake release function is only functional when the system is powered on.

3. Product Introduction

3.1. Product Name

Automated material handling robot

3.2. Product Variants

S4-1.5-0.4-Z0.4，S4-1.5-0.6-Z0.6，S4-1.5-0.6-Z0.8

3.3. Features

The LabGenius S4 automated material handling robot is a high-speed, precise, safe, reliable and easy-to-use robot developed by Bioyond Robotics.

Key Features:

- Compact design and lightweight design(20kg/44lbs)
- Extensive work range
- Fast and precise movement
- Enhanced safety during operation
- User friendly interface
- Flexible and versatile customer connectivity

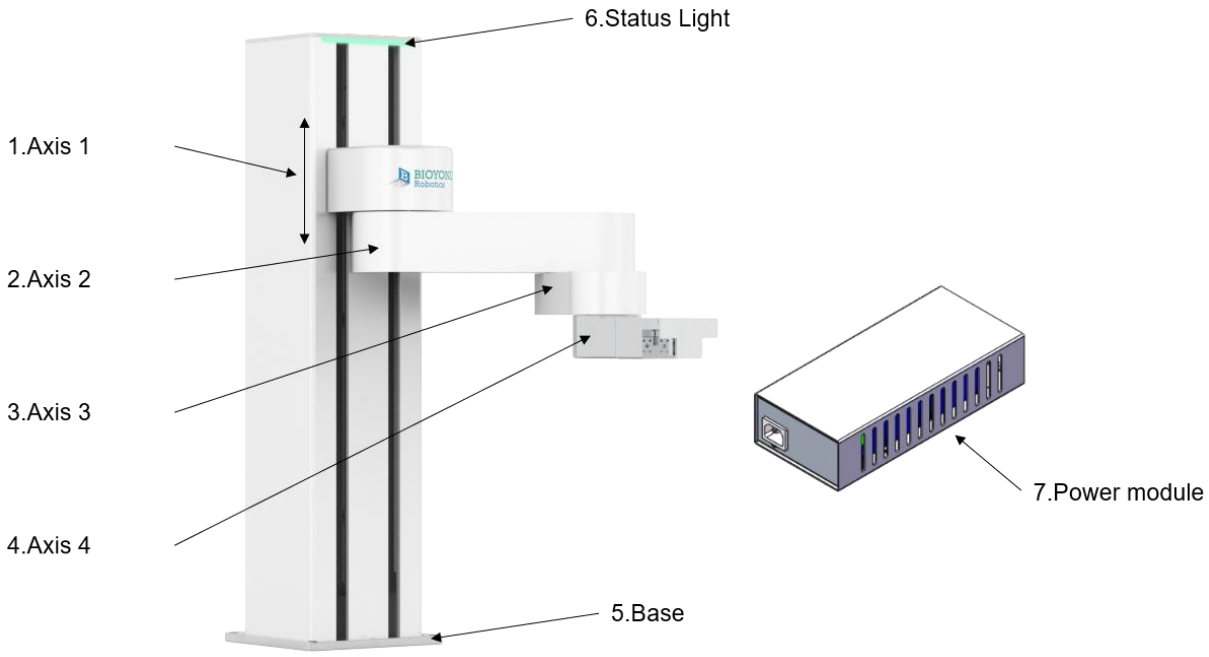
3.4. Technical Specification

Model	S4-1.5-0.4-Z0.4	S4-1.5-0.6-Z0.6	S4-1.5-0.6-Z0.8
Footprint	180×180mm	180×180mm	180×180mm
Reach	400 mm	600 mm	600 mm
Vertical travel	400 mm	600 mm	800 mm
Payload	1.5 kg	1.5 kg	1.5 kg
Stop time (When the emergency stop is triggered)	200ms		
Repeatability	±0.02 mm		

Cycle Time	1.4 s		
Max. TCP Speed	1400 mm/s		
Range of motion	Axis 1：0-400mm Axis 2：±95° Axis 3：±180° Axis 4：±220°	Axis 1：0-600mm Axis 2：±95° Axis 3：±180° Axis 4：±220°	Axis 1：0-800mm Axis 2：±95° Axis 3：±180° Axis 4：±220°
Weight	21kg	24kg	27kg
Options	Gripper, Flange, AI Camera, Barcode Scanner		
IO Interface	8×DI、8×DO		

3.5. Components

The LabGenius S4 automated material handling robot consists of a manipulator and a power module.



Additionally, the LabGenius S4 robot supports the following optional features:

- Force control gripper
- ISO-compliant flange
- Vision integration kit
- Scanner

3.6. Status Indicator

The status indicator on the top front of the LabGenius S4 robot provides at-a-glance system information. Various colors and illumination patterns indicate different operational states, such as standby, running, or fault pause:

Color	Robot Status
White	Power on, idle and not operational
Blue	System is enabled and prepared to start operating
Green	Robot is executing programmed operations
Yellow	Robot’s operation is temporarily halted
Red	System has detected a fault or error, requiring immediate attention

3.7. Option Tools

3.7.1.Force control gripper

Item	Parameter
Pay load	500g
Travel distance	28mm (Single side)
Repeatability	±0.1mm
Features	Force control, power down hold

3.7.2.ISO-compliant flange

The LabGenius S4 robot is equipped with an ISO flange for mounting third-party tools. For the mechanical interface of the flange, refer to section 4.1.3; for the electrical connections, refer to section 3.8.2.

3.7.3.Vision kits

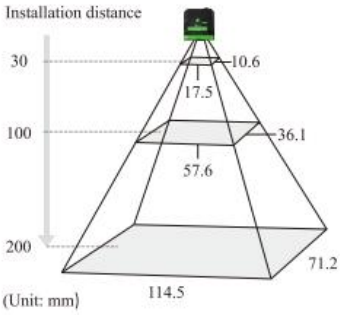
Item	Parameter
Effective pixel	630M pixel
Resolution	3072x2048
Max. frame rate	18.5fps
Communication	1000M Ethernet
Power supply	12-24V

3.7.4.Scanner

Item	Parameter
Barcode types	1D code: Code 39，Code 93，code 128，ITF25，ITF14， EAN8， EAN13，UPC_A，UPC_E，PHARMARCODE，CodaBar etc. 2D code: QR Code，Data Matrix，PDF417 etc.
Max. frame rate	50fps
Scan speed	60 code/s
Resolution	1280x800
Exposure time	20us~1s
Read distance	100mm
Communication	100M Ethernet
Power	24VDC±10%，maximum power consumption: 20W

Working parameters:

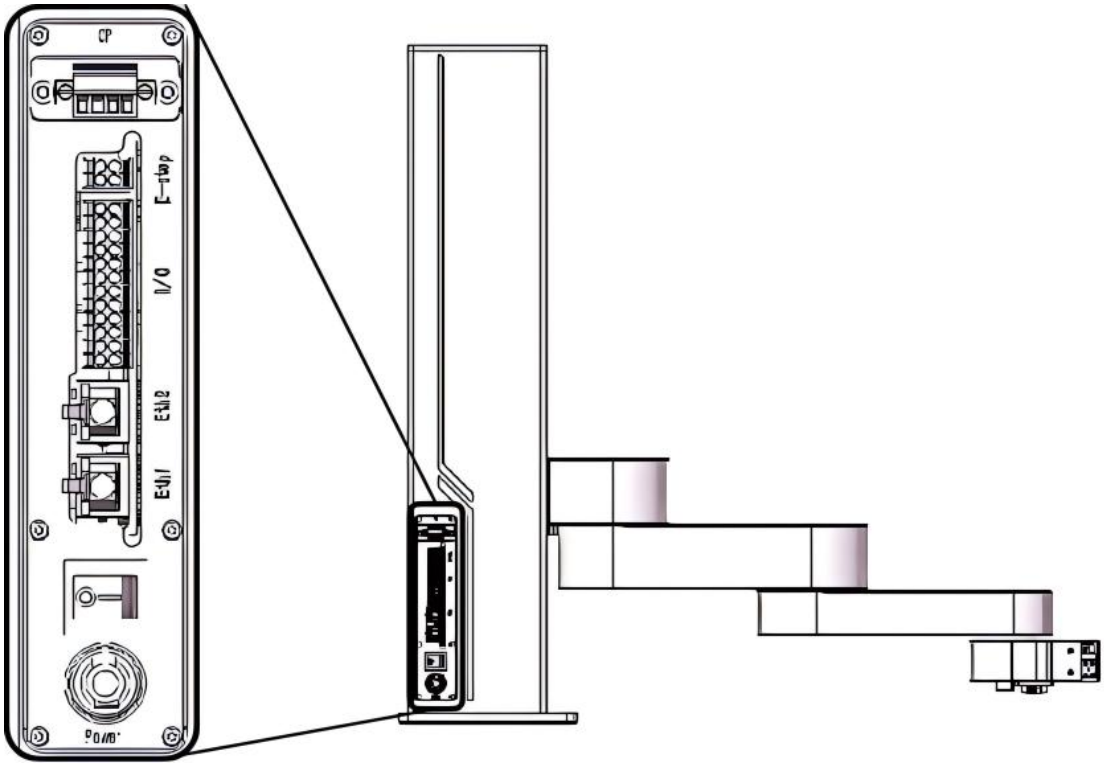
Lens focal length	Working distance (mm)	View(mm)		Minimum recognition accuracy(mm)	
		Horizontal	Vertical	QR code	One-dimensional code
6mm	30	17.5	10.6	0.055	0.027
	40	22.7	14.0	0.071	0.035
	70	41.3	25.7	0.129	0.065
	80	47.1	29.6	0.147	0.074
	100	57.6	36.1	0.180	0.090
	120	68.8	43.2	0.215	0.108
	140	80.2	50.4	0.251	0.125
	200	114.5	71.2	0.358	0.179



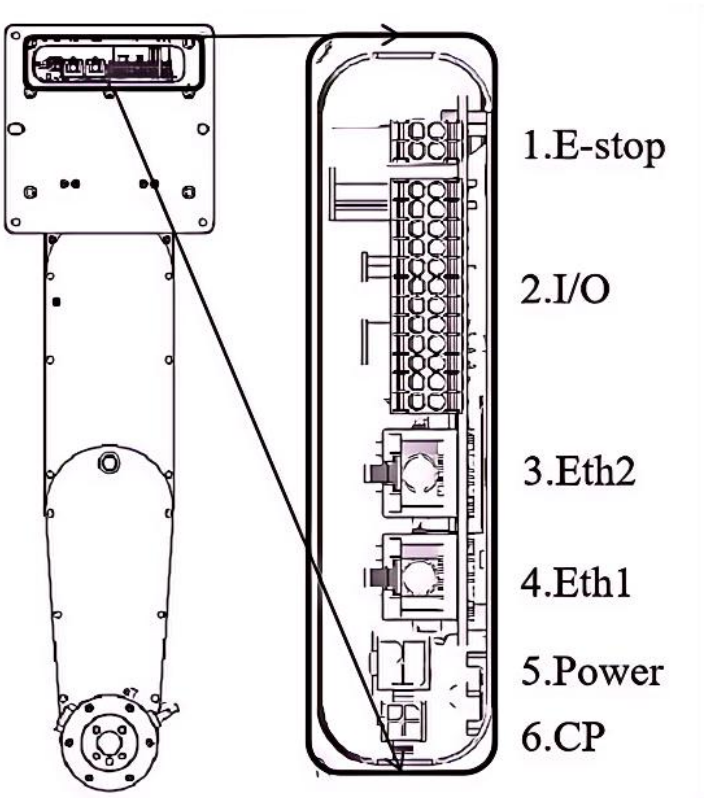
3.8. Electrical Connection

3.8.1. Robot Electrical Interface

- Lateral Wiring Configuration Port Diagram:

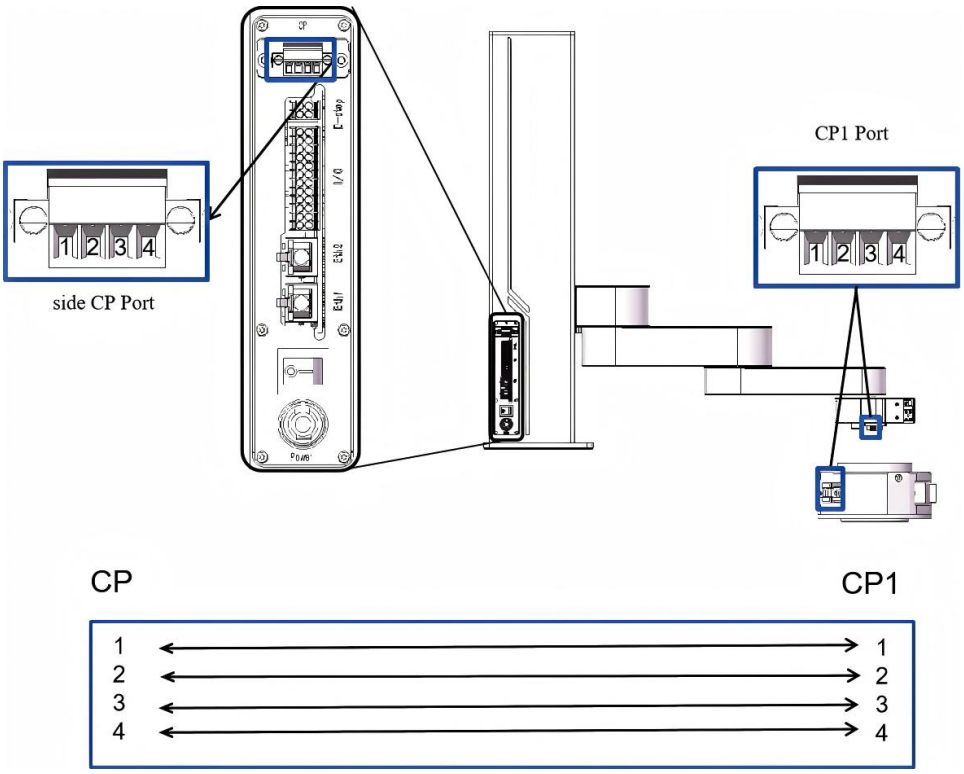


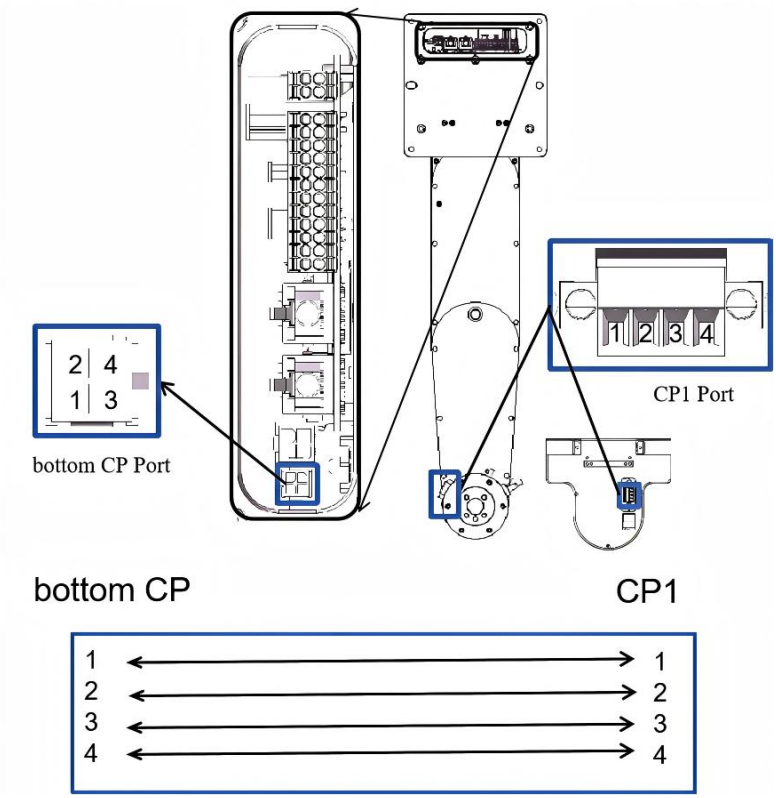
- Bottom Wiring Configuration Port Diagram:



3.8.2. Port Description:

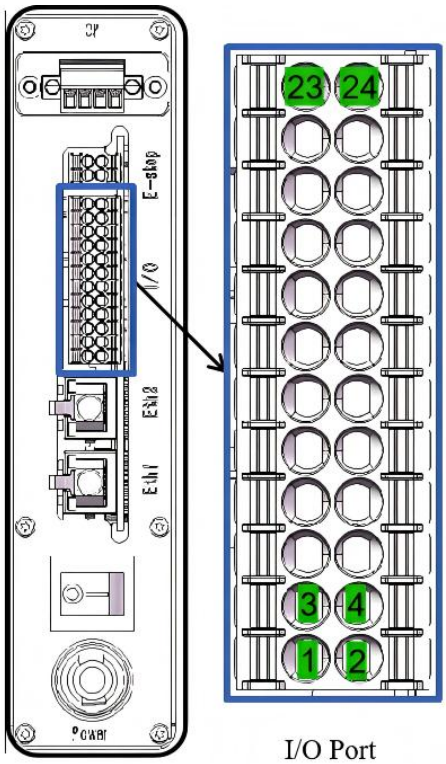
- CP port: application power port





The additional CP port is located at the bottom of the robot's arm. It extends the power port connection to the robot's panel port (CP1). The correspondence of the ports is illustrated in the figure above.

- E-Stop Interface: See section 2.6.1.
- I/O interface:



I/O Port

Signal definition of I/O interface：

Terminal	Signal	Terminal	Signal
1	0V_IO	2	24V_IO
3	0V_IO	4	24V_IO
5	DI[1]	6	DI[0]
7	DI[3]	8	DI[2]
9	DI[5]	10	DI[4]
11	DI[7]	12	DI[6]
13	DI_COM	14	DI_COM
15	DO[1]	16	DO[0]
17	DO[3]	18	DO[2]
19	DO[5]	20	DO[4]
21	DO[7]	22	DO[6]
23	DO_GND	24	DO_GND

Note：

24V_IO/0V_IO: The 24V power port of the robot, which can be used as the power supply for DI interface, or to power other devices. The parameters of 24V_IO port are as follows:

Parameter	Description
Nominal voltage	24V
Voltage tolerance	-5% ~ +5%
Rated output current	0.5A

The DI input interface can be configured as NPN mode or PNP mode. in NPN mode, DI_COM is

connected to the positive pole of the power supply, and DI[x] is connected to the signal input; in PNP mode, DI_COM is connected to the negative pole of the power supply and DI[x] is connected to the signal input.
The DI can utilize either 24V_IO/0V_IO as its power supply or an external independent 24V power source. The parameters of DI port are as follows：

Parameter	Description
Operating range	Refer to IEC61131-2, type 2
Maximum frequency	20Hz

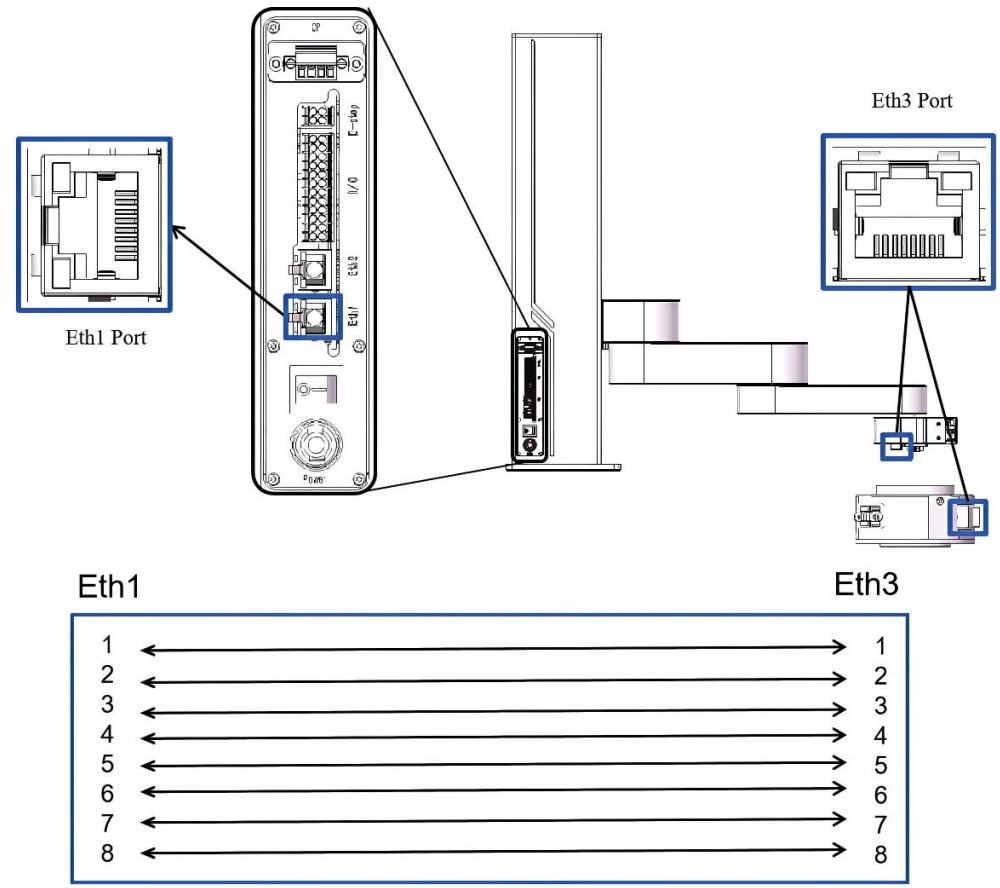
The DO is a 24V-compatible PNP-type output port that uses DO_GND as its reference terminal. The interface parameters are as follows:

Parameter	Description
Operating range	Refer to IEC61131-2, type 0.5
Maximum frequency	20Hz


：Although each DO port supports a current drive of 0.5A, the combined current of 24V_IO and all DO outputs must not exceed 1A due to the internal power supply limitations of the Lab Genius S4 Robot. Such as the following combination:

Example 1: 24V_IO 0.5A, all DOs: 0.5A
Example 2: 24V_IO 0.2A, all DOs: 0.8A

- ETH2: This is the standard port. Refer to section 4.3.3
- ETH1: This is a multi-functional application port. It extends the port at the end of the robot arm to the panel interface, pin definitions are application specific, and the correspondence is as follows:



Note: The circuit diagram uses lateral wiring option (Eth1) as an example, and the interface has the same connection configuration for bottom wiring option (Eth3).
The port can be connected using a standard Ethernet cable.

- Power interface

Connect the power module to the robot using the power cord provided in the packaging.

3.9. Prevailing Standards and Directives

- Machinery Directive 2006/42/EC
- EN ISO 10218-1:2011 Robots for Industrial Environments, Safety Requirements
- EN 60204-1:2016 Safety of Machinery, Electrical Equipment of Machines
- UL1740:2018 Standard for safety, Robots and Robotic Equipment
- CSA Z434: Industrial Robots and Robot Systems - General Safety Requirements
- ISO/TS15066:2016 Robots and robotic devices — Collaborative robots
- EN ISO 13849-1:2023Safety of machinery — Safety-related parts of control systems

3.10. Environmental Requirements

3.10.1. Transportation and Storage

This equipment must be transported and stored under the following environmental conditions:

- Ambient temperature: -40°C - 70°C.
- Relative humidity: 10% to 90%.
- Altitude: below 2,000 meters.

The transportation and storage environment for the packaged equipment must be dry, well-ventilated, and free from direct sunlight and corrosive gases. The equipment must be positioned in the specified orientation, and stacking should not exceed three layers.

When moving the equipment over short distances (e.g., between rooms within the same building), please adhere to the following precautions to ensure safety for both personnel and the equipment:

- Switch off the system power supply.
- Detach all peripheral components from the system.
- Ensure that no loose items are present on the control panel.
- Do not lift the equipment manually during transportation.
- Firmly hold the base of the machine for safe handling.
- Take measures to minimize the effects of vibration on the equipment.

For long-distance transportation of the equipment, please contact the manufacturer for confirmation and detailed guidance.

	After transportation, condensation may occur inside the equipment due to temperature differences between indoor and outdoor environments. Therefore, prior to powering the equipment, allow it to acclimatize to the new environment. If the ambient temperature is below 10°C, place the system in the operating environment for at least 30 minutes before use. For every additional 2.5°C in temperature difference, add an extra hour of acclimatization time.

3.10.2. Operating Environment

This equipment is designed to operate under the following environmental conditions:

- Power supply: AC 100 - 240V, 2A, 50/60Hz.
- Ambient temperature: 10°C - 40°C.
- Relative humidity: 20% to 80%. When the temperature is below 31°C, the maximum relative humidity can reach 80%; when the temperature rises to 40°C, the relative humidity linearly decreases to 50%.
- Altitude requirement: below 2,000 meters.
- Rated pollution level: Pollution level I, indicating no pollution or only dry, non-conductive contaminants are present.

	The equipment must be installed in an environment free from strong electromagnetic field interference, well-ventilated, dry, non-toxic, and free of corrosive gases. Additionally， it should be protected from direct sunlight and rain.
	It is mandatory to connect the equipment to a power socket with proper protective grounding. The use of a three-prong to two-prong power adapter is **strictly prohibited**. If the power socket lacks protective grounding, an additional reliable grounding terminal must be provided.

Never operate the equipment in the following hazardous conditions:

- Environments containing flammable, explosive, or toxic gases.
- Areas with fog or water splashing.
- Locations subject to severe impacts or vibrations.
- Spaces exposed to direct sunlight.
- Areas with significant fluctuations in AC voltage.

	To ensure proper ventilation, maintain a clearance of 20 - 30 cm between the rear of the equipment and the wall or adjacent objects. Failure to do so may result in increased internal temperature and potential malfunctions. Unauthorized disassembly and assembly of the equipment are not allowed, and any resulting consequences will be the sole responsibility of the user.

ESD Precautions:

- To prevent electrostatic build-up, store, maintain, and operate the equipment in an environment with a relative humidity of at least 30%.
- To protect the equipment from potential electrostatic discharge, always touch the metal frame of the components or a nearby large metal object before handling the equipment.

4. Installation and Quick Start

4.1. Preparation for Installation



: Please read this section carefully and install the equipment according to the requirements before proceeding with power-on and start-up operations.

4.1.1. Robot Installation Requirement

Please ensure that the installation base prepared for the S4 meets the following conditions:

- The base must be stable, non-wobbly, and level. Tables or workbenches with wheels, even those with locks, are not recommended. The flatness of the base surface should meet a minimum requirement of 0.1/500mm.
- The tilt angle of the base should not exceed 5°.
- The minimum natural frequency of the base should be no less than 22Hz. A lower natural frequency may adversely affect the lifespan of the robotic arm.

4.1.2. Packing List

The LabGenius S4 robot's package includes a cardboard box, foam padding, a dust-proof bag, the S4 robotic arm, a power module, and related accessories. The accessories consist of the following: an instruction manual, CP base installation connector, CP end-effector installation connector, E-Stop connector, I/O connector, power adapter, flash drive and AC power cable, and other components.

- LabGenius S4 automated material handing robot packing list (Gripper configuration)：

	Item	Amount
1	Manipulator (including a dust-proof bag)	1
2	Power Module	1
3	Power Transfer Cable	1

4	AC power cable	1
5	CP connector (for side outlet)	1
6	CP connector + contact (for bottom outlet)	1
7	CP connector (for end-effector outside)	1
8	E-Stop module	1
9	I/O connector	1
10	Standard gripper	1
11	Flash drive (including LabGenius S4 manual, software)	1
12	Certificate Card	1
13	Warranty Card	1

- LabGenius S4 automated material handing robot packing list (ISO flange configuration)：

	Item	Amount
1	Manipulator (including a dust-proof bag)	1
2	Power Module	1
3	Power Transfer Cable	1
4	AC power cable	1
5	CP connector (for side outlet)	1
6	CP connector + contact (for bottom outlet)	1
7	CP connector (for end-effector outside)	1
8	E-Stop module	1
9	I/O connector	1
10	ISO flange module	1
11	Flash drive (including LabGenius S4 manual, software)	1
12	Certificate Card	1
13	Warranty Card	1

The camera module package includes a cardboard box, foam padding, a dust-proof bag, camera unit and related accessories. The accessories consist of the following: Ethernet cable, screw, certificate card , and other components.

- Camera module packing list (ISO flange configuration)：

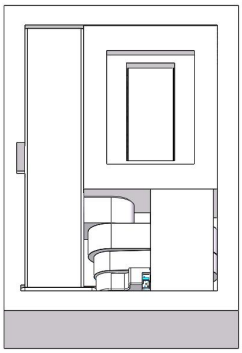
	Item	Amount
1	Camera unit (including a dust-proof bag)	1
2	Ethernet cable	1
3	Screw	1
4	Certificate Card	1
5	Warranty Card	1

The scanner module package includes a cardboard box, foam padding, a dust-proof bag, scanner unit and related accessories. The accessories consist of the following: screw, Certificate Card, and other components.

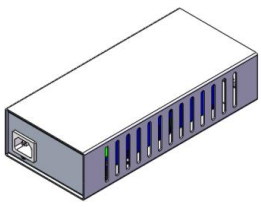
- Scanner module packing list (ISO flange configuration)：

	Item	Amount
1	Scanner unit (including a dust-proof bag)	1
2	Screw	1
3	Certificate Card	1
4	Warranty Card	1

The package is shown below:



Additionally, a power module is provided for the manipulator:

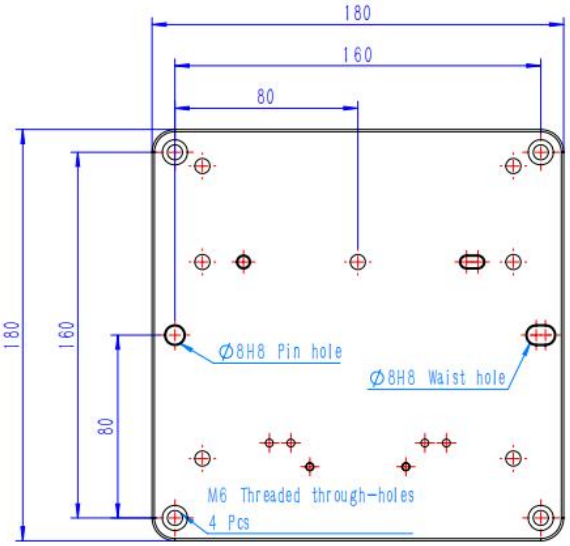


The robotic arm’s power module measures 244mm x 111mm x 57mm and connects to the robotic arm

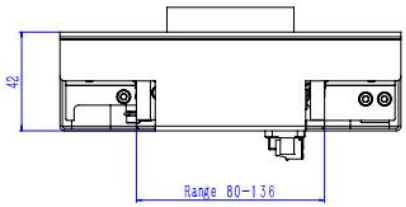
body using the power adapter cable.

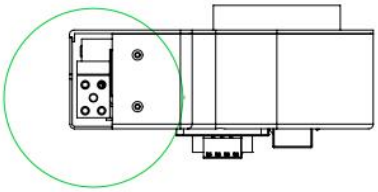
4.1.3.Robot Mechanical Interface

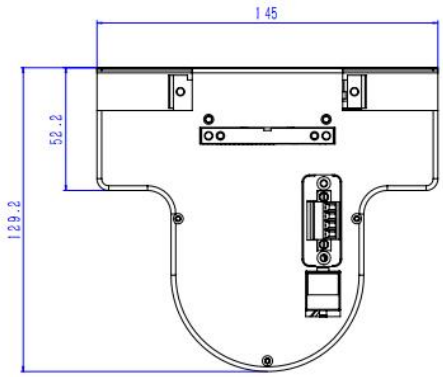
Robotic Arm Base:

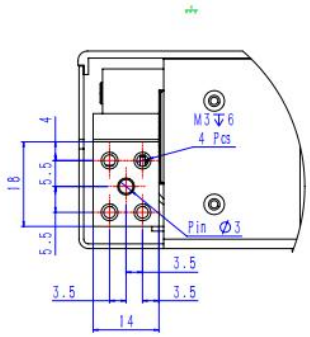


Robotic Arm End-Effector Gripper Interface:

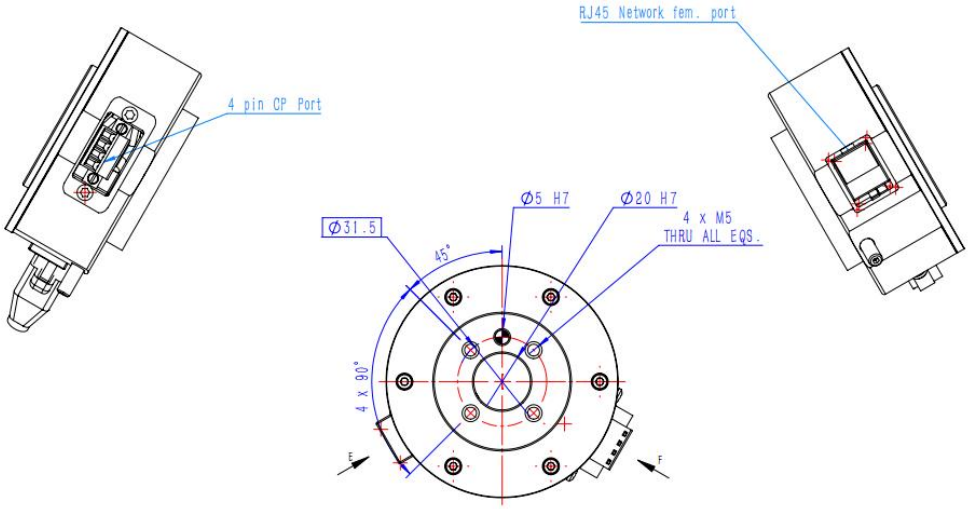




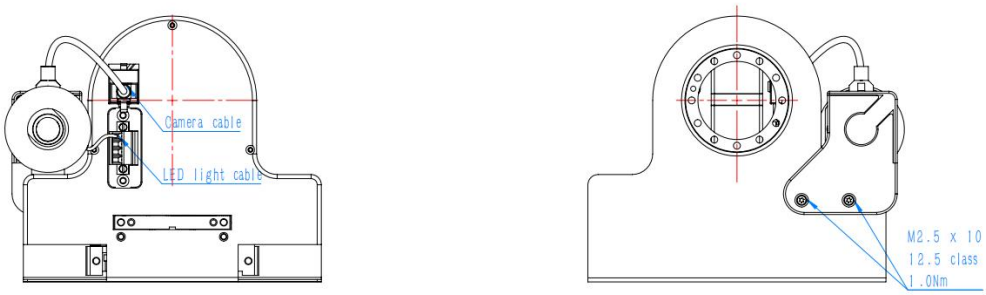




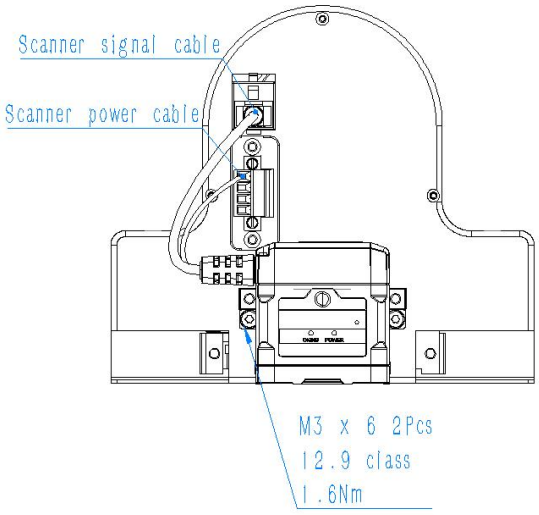
Robotic Arm End-Effector ISO Flange Interface:



The Camera Module Interface：



The Scan Module Interface：



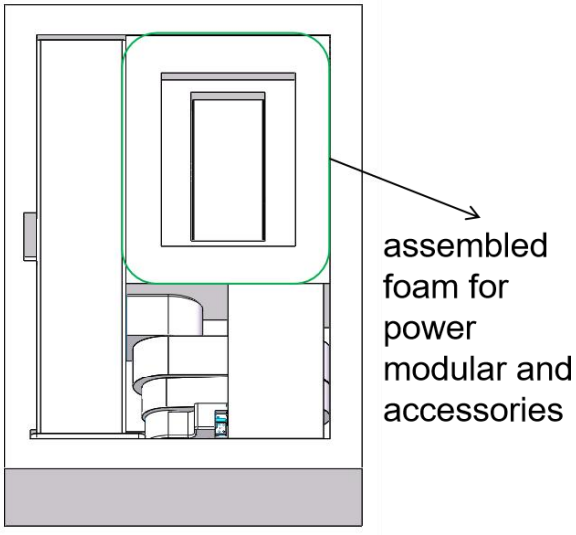
4.2. Unpacking and Installing

: Personnel operating the LabGenius S4 must have the required knowledge of mechanical and electrical installation/maintenance/repair and comply with applicable national/local laws and regulations.

: When opening the LabGenius S4 packaging, the operator must check whether the equipment and accessories are complete, and verify they are free from damage.

Unpacking steps:

- Step 1: inspect the exterior of the cardboard box for any signs of damage. If any damage is detected, promptly trace the logistics record and contact us.
- Step 2: Remove the tape from the top of the cardboard box, open the box, and take off the top layer of foam padding to reveal the robotic arm body, power module, and related accessories.
- Step 3: Carefully remove the foam padding securing the power module and related accessories.

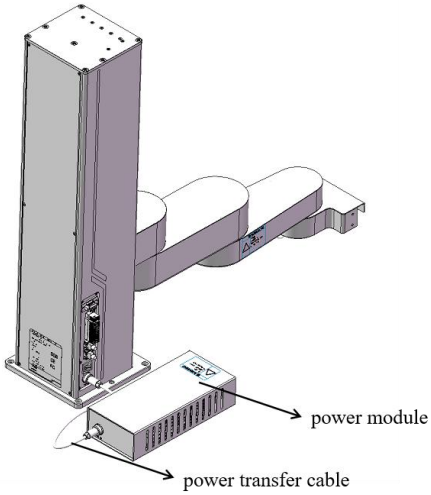


- Step 4: Carefully remove the S4 robotic arm from the foam padding and move it to the final assembly location.
- Step 5: Remove the foam padding securing the upper and lower arms.
- Step 6: Secure the S4 robotic arm to the base by installing the fasteners according to specifications in the table below:
-

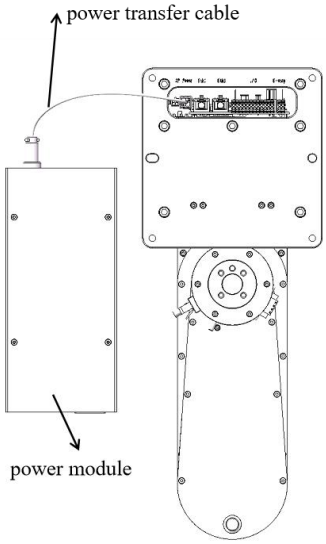
	Specification	Requirement
Screw	4 PCS M6 x 25	11Nm (12.9 class)
Pin	⌀8mm Pin	GB /T 119.1 - 2000

- Step 7： Install the E-Stop connector onto the robotic arm body as described in section 2.6.1. Connect one end of the power adapter cable to the power module and the other end to the robotic arm body. Finally, connect the AC power cable to the power module, completing the unpacking and installation process.

Lateral port diagram:



Bottom port diagram:



(Note: It is recommended to have a partner assist during the unpacking and assembly process. The estimated completion time is approximately 10 minutes.)

Before unpacking, inspect the packaging for any signs of damage or abnormalities. As shown in the image below, open the cardboard box and remove the top layer of foam padding to reveal the S4 robotic arm body.

4.3. Quick Start


：The LabGenius S4 Robot requires configuration through Bioyond Works to operate.

4.3.1. Software Installation

Bioyond Works software requires a Windows 10 or higher operating system. An installer is provided, follow these steps to install Bioyond Works:

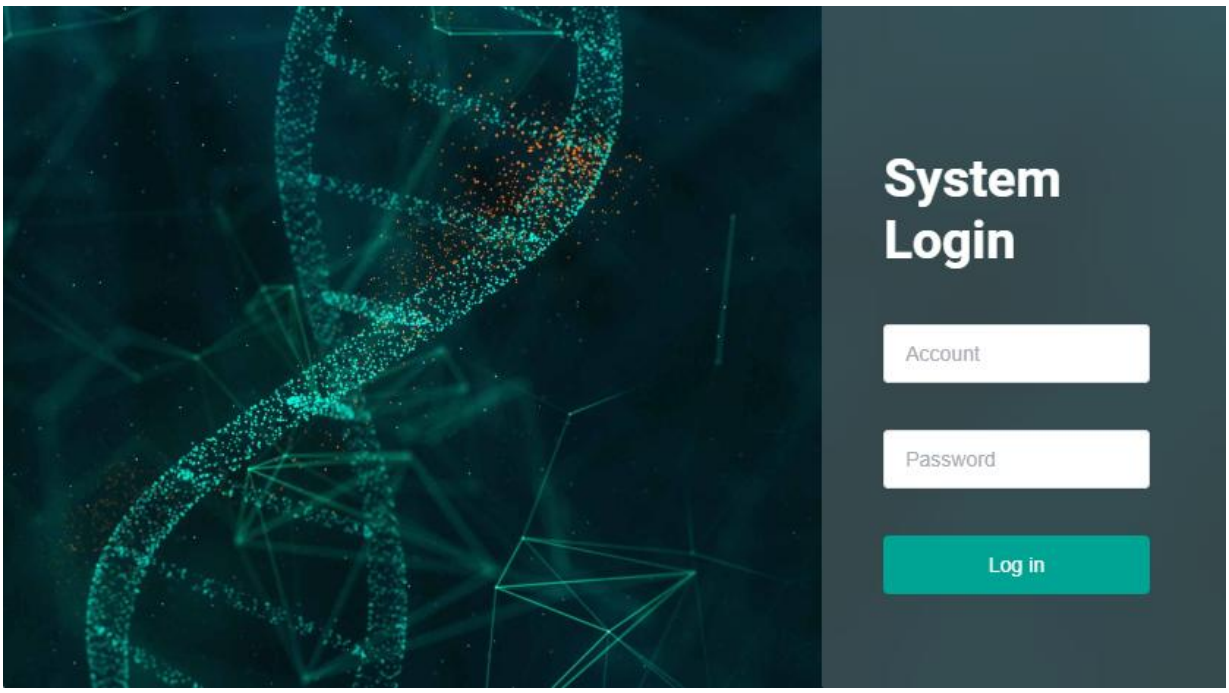
- Open the Bioyond_Works_Robot.exe file.
- Follow the installer instructions to complete the installation process. After the installation is complete, the application will be launched automatically. Grant the application security or firewall permissions, If prompted by the system, to ensure Bioyond Works can start and communicate with the controller over the network.

4.3.2. Power On

- **Lateral Wiring Configuration:** Turn on the main power switch on the side of the robotic arm's pillar to power on the robot.
- **Bottom Wiring Configuration:** The robot powers on automatically when the external power supply is turned on, requiring no further actions.

4.3.3. Software Startup and Login

Double-click the Bioyond Works icon to launch the software. Upon launch, the user login page will appear. Enter your credentials to access the software homepage.

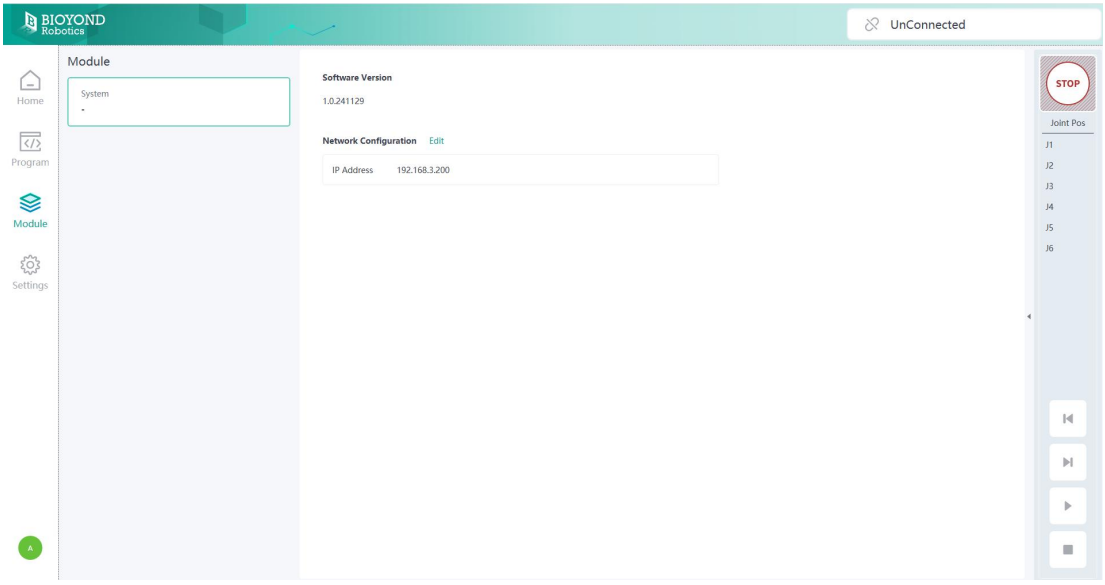


4.3.4.Connection

Bioyond Works software supports connecting to the material handling robot via both wired (Ethernet) and wireless (Wi-Fi) methods. The specific steps for establishing a connection are as follows:

Wired connection (Ethernet):

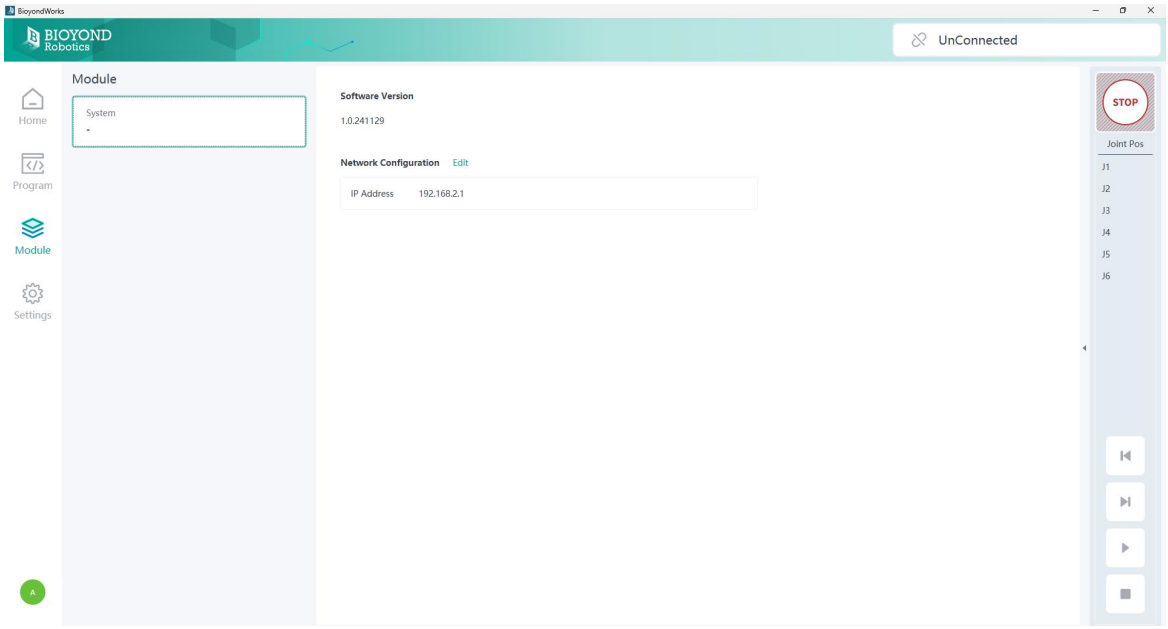
- Connect the computer with Bioyond Works installed to LabGenius S4 using an Ethernet cable.
- Set the computer's IP address to 192.168.3.x(it must not match the controller's IP).
- Open the Bioyond Works, select “Modules” in the menu bar, set the IP address to 192.168.3.200 in the network configuration page, as shown in the following figure
- Bioyond Works will communicate with LabGenius S4 automatically.



Wireless connection (Wi-Fi):

- Select the LabGenius S4 from the computer's WiFi list.

- Open the Bioyond Works, select “Modules” in the menu bar, set the IP address to 192.168.2.1 on the network configuration page, as shown in the following figure.
- Bioyond Works will communicate with LabGenius S4 automatically.



4.3.5.New and Execution Program

A pre-configured robot demo process is stored within the robot controller. Upon a software startup, it will automatically retrieve the programs from the robot controller. Follow these steps to execute a program:

1) Edit and program

Open the main program **main. Lua** files in the software's programming page. Edit the existing subprograms or add new motion instructions as needed

2) Adjust motion speed

To ensure safety during the robot's motion, adjust the **speed slider** in the **Jog Control** page to reduce the robot's movement speed.

3) Enable the controller

Click the **Enable** button to set the controller to the servo-enabled state, preparing the robot to execute tasks.

4) Run the program

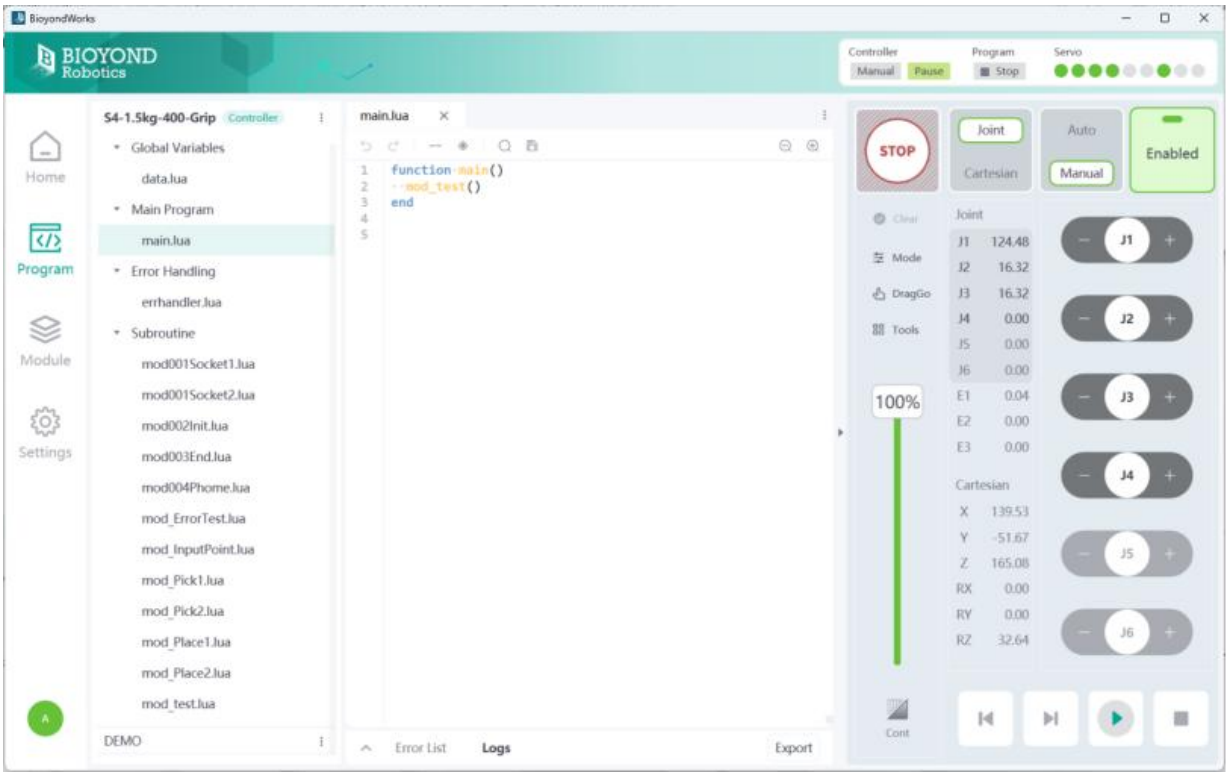
Click the **Run** button to start the program. Once the program starts, the button will change to **Pause**. The real-time execution status, including the currently running subprogram filename and the executed line number, will be displayed at the status bar in the top-right corner of the software.

5) Pause and stop the program

- **Pause the Program**: Click the **Pause** button to halt the program execution.
- **Stop the Program**: Click the **Stop** button to terminate the current program.

6) Stop

For critical stops or controller interruptions, a software **Stop** button is available. Activating **Stop** button will halt robot motion through software controls, disable the motor, and bring the system to a controlled stop. Once the stop condition has been resolved or the system is ready to resume, press the **Stop** button again to restore normal operation.



5. Maintenance and Service

This chapter outlines detailed requirements for maintenance and serving the equipment to ensure normal operation, as well as methods for troubleshooting common issues. Although the equipment undergoes strict quality inspections before leaving the factory, most issues may result from sudden changes in the operating environment, accidental loosening of connectors, or incomplete system setup. For faults that do not require disassembling, users can refer to the troubleshooting table and follow the recommended solutions or contact our customer service department. For faults requiring disassembly, users must contact our customer service department for repairs; self-repair is strictly prohibited.

If defects due to manufacturing or assembly are identified during initial use, Bioyond Robotics will provide spare parts for replacement or repair of affected components. All replaced parts or equipment should be returned to Bioyond Robotics. If the warranty period has expired, Bioyond Robotics reserves the right to charge the customer for replacement parts or repair services. Bioyond Robotics does not take liabilities for any losses caused by equipment defects, such as production downtime or damage to other equipment.

5.1. Equipment Maintenance

Regular maintenance and upkeep are essential for safe and reliable operation of the equipment. Proper care minimizes downtime, reduces service costs, and mitigates operational hazards. Before performing any maintenance or upkeep, the equipment must be powered off and disconnected from the power supply.

5.1.1. Safety Checks

Before each use:

- Inspect all cables and power lines for cracks, cuts, or abrasions.

Conduct the following monthly checks:

- Inspect the equipment for any mechanical issues.
- Check for any missing or loose hardware.

- Ensure that all connection cables are securely fastened and not loose.
- Ensure Emergency stop is functioning.

5.1.2. Equipment Cleaning

Weekly cleaning instructions:

- Outer Surfaces and Cables: Wipe the outer surfaces with a soft, clean, and dry cloth. For stubborn dirt, use a cloth lightly dampened with a mild detergent.

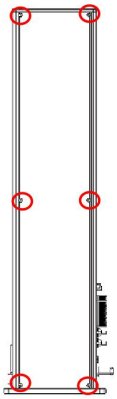
5.1.3. Battery Replacement

The LabGenius S4 automated material handling robot contains a battery for recording critical data. Bioyond Robotics recommends turning on the robot at least once every month to check for any low voltage battery alarms. If a warning is displayed, replace the battery immediately. Even if no warning appears, Bioyond Robotics recommends replacing the battery at least once every 3 years as a preventive measure.

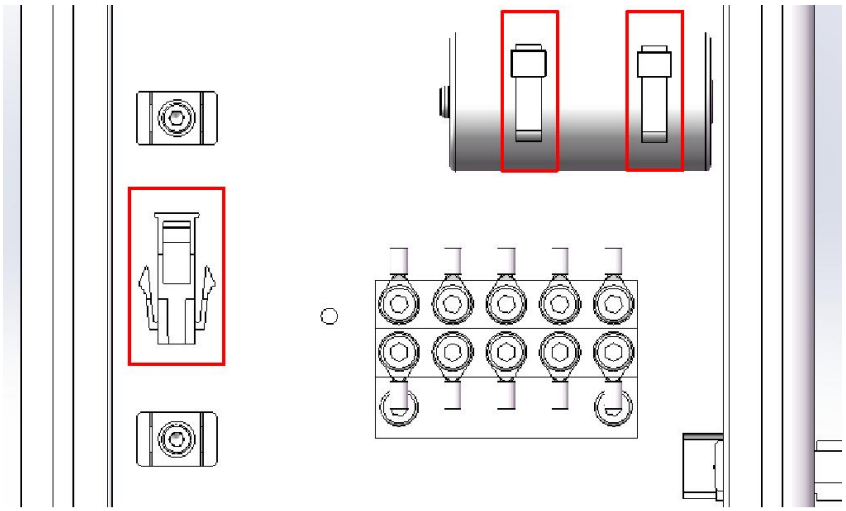
The battery spare part number is 1.02.10.10016. Please contact Bioyond Robotics for assistance.

Steps for replacing the battery:

1. Lead the Robot to the Home Position: Refer to related instructions to guide the robot to the home position.
2. Turn off robot power.
3. Remove the Rear Cover Plate of the Robot Column: Use an M2.5 screwdriver to remove the screws (as indicated in the diagram below) and detach the rear cover plate from the column.



4. Cut the Battery Retaining Strap: Use scissors to carefully cut the strap securing the battery, then disconnect the battery connector.



5. Connect the New Battery: Plug the new battery into the connector and secure it in its original position using a retaining strap.
6. Reattach the Robot's Rear Cover Plate: Use an M2.5 screwdriver, reinstall the screws to firmly secure the rear cover plate onto the column.

5.2. Software Updates

The software is regularly updated to enhance functionality, improve user experience, and address any identified issues. Users will be notified of new updates, which will include detailed information about the changes and improvements.

Additionally, the software provides monitoring applications with log tracking information, recording operational details when errors occur. These logs are available to maintenance or repair personnel for troubleshooting and diagnostics.

5.3. Troubleshooting

Although the equipment undergoes rigorous quality checks before leaving our facility, faults may still occur due to factors such as user operation, handling, and environmental conditions. Issues such as user interface errors, application logic errors, or system/network resource availability problems may cause the software to fail, requiring a restart. After restarting, the software will resume normal operation, and no data will be lost even in the event of an unexpected power outage.

Critical software defects, such as unexpected crashes, can often be resolved by restarting the software. The troubleshooting table below provides a structured approach to identifying and resolving various faults by listing their symptoms, potential causes, and recommended solutions.

For issues that can be resolved without disassembling the device, users are encouraged to consult the troubleshooting table and follow the recommended solutions. If the issue persists or requires disassembly, users must contact our customer service department for assistance; self-repair or unauthorized disassembly is prohibited.



：If the LabGenius S4 requires repairs due to a fault, Bioyond Robotics can provide circuit diagrams, component lists, or other necessary information to qualified service personnel upon request.

Troubleshooting Table

Fault Symptoms	RCA	Solution
Equipment not functioning	1、Power switch not turned on	1、Press the "Power" button
	2、Not connected to the power supply	2、Connect to the power supply
	3、Power cable damaged	4、Replace the power cable
	4、Switch malfunction	5、Contact the manufacturer for repair
Unable to provide normal prompts	1、Incorrect prompt settings	1、Check the device settings
	2、Prompt errors	2、Contact the manufacturer for repair
System software fails to start or operate	A virus has infiltrated the system, causing system errors	Contact the manufacturer for repair



Caution: The internal power supply is a non-serviceable part and must not be disassembled.

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5.4. Disposal of Equipment and Accessories at the End of Their Service Life

All used grease, lubricants and batteries must be disposed of in accordance with the applicable laws and regulations in your country.

If part or all the equipment needs to be disposed of, the individual parts must be sorted by material type (e.g., grouping all ferrous parts together and all plastic components together) and disposed of accordingly. Disposal must comply with the applicable laws and regulations in the country or region.

The LabGenius S4 is composed of components made from various materials. At the end of its life cycle, all materials should be properly disassembled, recycled or reused in compliance with applicable laws and industry standards. Reusing or recycling equipment and parts helps to conserve natural resources and reduce environmental impact.

- WEEE Label

The following symbol indicates that the product must not be disposed of as regular household waste. Each product must be handled in accordance with local regulations applicable to its specific materials and components.

Bioyond Robotics is committed to complying with the EU Directive on Waste Electrical and Electronic Equipment (WEEE-2012/19/EU). Our objective is to ensure that our products are properly disposed of or recycled at the end of their useful life. Products from Bioyond Robotics that fall under the WEEE directive are marked with the following Crossed-out Wheelie Bin symbol, indicating that they should not be disposed of with regular household waste but must be collected and handled separately. If you or your business possess Bioyond Robotics's products that have reached the end of their life or require disposal for another reason, please contact Bioyond Robotics for assistance with proper disposal and recycling.



- China RoHS symbol



The symbol above provides information about hazardous substances and the environmental protection use period of LabGenius S4 in accordance with the "Management Methods for the Restriction of the Use of Hazardous Substances in Electrical and Electronic Products (SJ/T 11364-2014) ".
Orange symbol with a number: This indicates that the product contains certain hazardous substances but can be used safely during its environmental protection use period (denoted by the number in the center). After this period, the product should be entered into the recycling system.


: *This form and environmental protection use period label comply with the regulations in China. These are not applicable or required in other countries.*

- Lubricants and greases

All lubricants and greases must be disposed of by authorized personnel or contractors in compliance with local regulations. Do not dispose of lubricating oils and greases near bodies of water such as lakes, ponds, or ditches, nor discharge them into sewers or soil. If incineration is required, it must be performed under control conditions as mandated by local regulations and environmental standards.

Appendix 1: Certifications



SGS

SGS C US

CERTIFICATE OF COMPLIANCE

Certificate Number: SGSNA/24/SH/00483

Contract Number: 803011
Certificate Project Number: SH-CERT240905662

Certified Product: Automated Material Handling Robot
Trademarks: BIOYOND Robotics

Model(s): S4-1.5-0.4-Z0.4, S4-1.5-0.6-Z0.6, S4-1.5-0.6-Z0.8
Technical Data: 100-240 V~; 50/60 Hz;
S4-1.5-0.4-Z0.4: Payload 1.5 kg; Max. reach 0.4 m;
S4-1.5-0.6-Z0.6: Payload 1.5 kg; Max. reach 0.6 m;
S4-1.5-0.6-Z0.8: Payload 1.5 kg; Max. reach 0.6 m

Certificate Holder: Bioyond Robotics (Shanghai) Co., Ltd.
No. 101, 109, 110, Building 1, 207 Huanqiao Road, Pudong New District, Shanghai, 201315, China

This certificate supercedes previous certificates issued with the same certificate number. Certification is valid when products are indicated on the SGS directory of certified products at www.sgs.com or using the QR code below. The product is certified according to ISO/IEC Guide 17067, Conformity assessment - Fundamentals of product certification, System 3, and in accordance with:

UL 1740 4th Edition, Revised August 11, 2023
ANSI/RIA R15.06-2012
NFPA 79-2024, Dated December 21, 2023
CAN/CSA Z434-14 (R2019), Reaffirmed 2019

Authorized by:

Mark Lohmann
Certifier

Effective date: 02 December 2024

Page 1 of 1

SGS operates certification programs under the authority of several accreditation or recognition bodies including A2LA, ANAB, OSHA NRTL, and Standards Council of Canada. This certificate is issued by the company under its General Conditions for Certification Services accessible at https://www.sgs.com/terms-and-conditions. Attention is drawn to the limitations of liability defined therein and in the Test Report here above mentioned which findings are reflected in this Certificate. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.

Certification Body

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VERIFICATION OF MD COMPLIANCE

No.:	MD SHES2408016254MD
Applicant:	Bioyond Robotics (Shanghai) Co., Ltd. No. 101, 109, 110, Building 1, 207 Huanqiao Road, Pudong New District, Shanghai, 201315, China
Manufacturer:	Bioyond Robotics (Shanghai) Co., Ltd. No. 101, 109, 110, Building 1, 207 Huanqiao Road, Pudong New District, Shanghai, 201315, China
Product Description:	Automated Material Handling Robot
Model No.:	S4-1.5-0.4-Z0.4, S4-1.5-0.6-Z0.6, S4-1.5-0.6-Z0.8.
Trade Mark:	BIOYOND Robotics
Additional Information (if any):	--

Sufficient samples of the product have been tested and found to be in conformity with requirements of following standard(s) and applicable parts of the Annex I of the Machinery Directive (2006/42/EC)

Test Standard:	EN ISO 10218-1:2011 EN 60204-1:2018 EN ISO 12100:2010
as shown in the Test Report Number(s):	SHES240801625401-01/02/03/04

This Verification of MD Compliance has been granted to the applicant based on the results of tests, performed by Laboratory of SGS-CSTC Standards Technical Services Co., Ltd. on sample of the above-mentioned product in accordance with the provisions of the relevant harmonized standards under the Machinery Directive 2006/42/EC. The CE mark can be affixed, under the responsibility of the manufacturer, after completion of an EC Declaration of Conformity and compliance with all relevant EC Directives. The affixing of the CE marking presumes in addition that the conditions in the Directives are fulfilled.

Andrew Zhai
Technical Manager
SGS-CSTC Standards Technical Services Co., Ltd.

2024-11-11

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Appendix 2: Spare Parts List

The LabGenius S4 spare parts include components compatible with the following variants:

- All configurations of S4-1.5-0.4-Z0.4
- All configurations of S4-1.5-0.6-Z0.6
- All configurations of S4-1.5-0.6-Z0.8

The LabGenius S4 spare parts are categorized into two levels: L1 and L2. Before conducting maintenance on any spare parts, it is crucial to verify the component level.

- L1 Level Spare Parts: L1 spare parts are designed for on-site replacement. Users must strictly adhere to the maintenance and replacement instructions provided in the relevant documentation. If any issues occur during replacement or maintenance, please contact customer service for assistance.
- L2 Level Spare Parts: Replacing L2 components requires specialized training and may require use of special tools. Only Bioyond Robotics's on-site maintenance personnel or individuals who have been trained and certified by Bioyond Robotics are authorized to perform the replacement of L2-level components.

No.	spare Part No.	Description	Category	Class
1	1.02.10.10016	Battery Module	Electrical comp.	L1
2	1.03.15.10008	RIB Board	Electrical comp.	L2
3	1.02.02.10002	6A Drive Board	Electrical comp.	L2
4	1.02.02.10003	10A Drive Board	Electrical comp.	L2
5	1.03.01.10002	Brake resistance	Electrical comp.	L2
6	1.02.05.10008	Switch	Electrical comp.	L2
7	1.02.10.10015	Brake Button Module	Electrical comp.	L2
8	2.01.03.00024	Lamp Module	Electrical comp.	L2
9	2.01.03.00002-01	Power Module	Electrical comp.	L1
10	2.01.03.00009-01	Power Wire-BT-0.5m	Electrical comp.	L1
11	2.01.03.00009-02	Power Wire-SD-0.5m	Electrical comp.	L1
12	2.01.03.00011	A1 Upside Pulley Module	Driver chain comp.	L2
13	2.01.03.00012	A1 Lower Pulley Module	Driver chain comp.	L2
14	2.01.03.00013	A2 Gear Module	Driver chain comp.	L2
15	2.01.03.00014	A2 End Pulley Module	Driver chain comp.	L2
16	2.01.03.00015	A3 Motor Module	Driver chain comp.	L2

17	2.01.03.00016	A3 Middle Pulley Module	Driver chain comp.	L2
18	2.01.03.00017	A3 End Pulley Module	Driver chain comp.	L2
19	2.01.03.00018	A4 Motor Module	Driver chain comp.	L2
20	2.01.03.00019	A4 Middle Pulley Module	Driver chain comp.	L2
21	2.01.03.00020	A4 End Pulley Module	Driver chain comp.	L2
22	2.01.01.00023	Gripper Module	Driver chain comp.	L1
23	2.01.03.00022	Scan Module	Options	L1
24	2.01.03.00023	Camera Module	Options	L1
25	2.01.03.00031	Camera Power Module	Options	L1
26	1.01.06.10097	Gripper Cover	Covers	L2
27	1.01.16.10032	Belt 246-3MGT-9	Driver chain comp.	L2
28	1.01.16.10015	Belt LL-5GT-15 1167mm	Driver chain comp.	L2
29	1.01.16.10015	Belt LL-5GT-15 1567mm	Driver chain comp.	L2
30	1.01.16.10015	Belt LL-5GT-15 1967mm	Driver chain comp.	L2
31	1.01.16.10033	Belt 426-3MGT-20	Driver chain comp.	L2
32	1.01.16.10034	Belt 390-3MGT-5	Driver chain comp.	L2
33	1.01.16.10035	Belt 408-3MGT-13	Driver chain comp.	L2
34	1.01.16.10037	Belt 333-3MGT-10	Driver chain comp.	L2
35	1.01.16.10036	Belt 375-3MGT-5	Driver chain comp.	L2
36	1.01.16.10046	Belt 600-3MGT-13	Driver chain comp.	L2
37	1.01.16.10047	Belt 366-3MGT-5	Driver chain comp.	L2
38	1.01.16.10048	Belt 540-3MGT-10	Driver chain comp.	L2
39	2.01.03.00034	E-stop Module	Electrical comp.	L1
40	1.02.07.10060	CP connector	Electrical comp.	L1
41	1.02.07.10064	IO connector	Electrical comp.	L1